

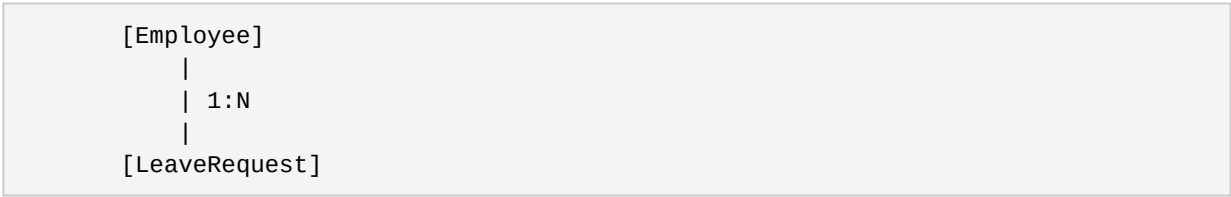
Data Model Documentation

Overview

The employee leave application manages data related to employees and their leave requests. It ensures that leave requests are submitted, processed, and approved in an organized manner. The system manages the following core domains:

- **Employee:** Responsible for storing employee details and their leave requests. The cardinality is 1:N, meaning one employee can have multiple leave requests.
- **LeaveRequest:** Responsible for storing details of each leave request, including the employee who requested it, the dates, and the status. The cardinality is 1:N, meaning one leave request is associated with one employee.

Entity Relationship Diagram



Entities

Employee

Purpose: Represents an employee in the organization.

Table/Collection Name: Employee

Fields:

Field Name	Type	Required	Constraints	Description
id	UUID	Yes	Primary key, auto-generated	Unique identifier for the employee
name	VARCHAR	Yes	Max 255 chars	Employee's full name
email	VARCHAR	Yes	Max 255 chars, unique	Employee's email address
created_at	TIMESTAMP	Yes	Auto-set on create	Record creation time

Field Name	Type	Required	Constraints	Description
updated_at	TIMESTAMP	Yes	Auto-set on create/update	Last modification time

Relationships:

- Has many LeaveRequest (1:N): Inverse via LeaveRequest.employee_id

Validation Rules:

- name must be non-empty
- email must be non-empty and unique across all records

LeaveRequest

Purpose: Represents a leave request made by an employee.

Table/Collection Name: LeaveRequest

Fields:

Field Name	Type	Required	Constraints	Description
id	UUID	Yes	Primary key, auto-generated	Unique identifier for the leave request
employee_id	UUID	Yes	Foreign key referencing Employee.id	Employee who requested the leave
start_date	DATE	Yes		Start date of the leave
end_date	DATE	Yes		End date of the leave
status	VARCHAR	Yes	Default 'Pending'	Status of the leave request
created_at	TIMESTAMP	Yes	Auto-set on create	Record creation time
updated_at	TIMESTAMP	Yes	Auto-set on create/update	Last modification time

Relationships:

- Belongs to Employee (N:1): Foreign key employee_id references Employee.id

Validation Rules:

- start_date must be a valid date
- end_date must be a valid date and must be after start_date
- status must be one of 'Pending', 'Approved', 'Rejected'

Relationships and Constraints

Foreign Keys

From Entity	Field	To Entity	Field	Cardinality	Delete Behavior
LeaveRequest	employee_id	Employee	id	N:1	CASCADE

Unique Constraints

- Employee(email): Ensures no duplicate email addresses

Data Validation Rules

Business Rules:

- start_date must be before end_date: If start_date is after end_date, return validation error.
- status must be one of 'Pending', 'Approved', 'Rejected': If status is not one of these values, return validation error.

Format Rules:

- email: Must match the regex `^[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}$`

Data Lifecycle

Creation

- A new LeaveRequest is created when an employee submits a leave request.
- Default status is Pending

- `created_at` is automatically set to the current timestamp

Updates

- `LeaveRequest.status` can only be updated by HR personnel
- `updated_at` is automatically updated on any modification

Archival / Soft Delete

- Records are not archived but can be marked as 'Rejected' or 'Approved' to indicate their status

Indexes

- Primary key: `id`
- Foreign keys: `employee_id`
- Search fields: `name`, `email` (for common queries)

Example Records

Employee

```
{
  "id": "550e8400-e29b-41d4-a716-446655440000",
  "name": "John Doe",
  "email": "john.doe@example.com",
  "created_at": "2026-01-15T10:30:00Z",
  "updated_at": "2026-01-15T10:30:00Z"
}
```

LeaveRequest

```
{
  "id": "123e4567-e89b-12d3-a456-426614174000",
  "employee_id": "550e8400-e29b-41d4-a716-446655440000",
  "start_date": "2026-01-15",
  "end_date": "2026-01-20",
  "status": "Pending",
  "created_at": "2026-01-15T10:30:00Z",
  "updated_at": "2026-01-15T10:30:00Z"
}
```